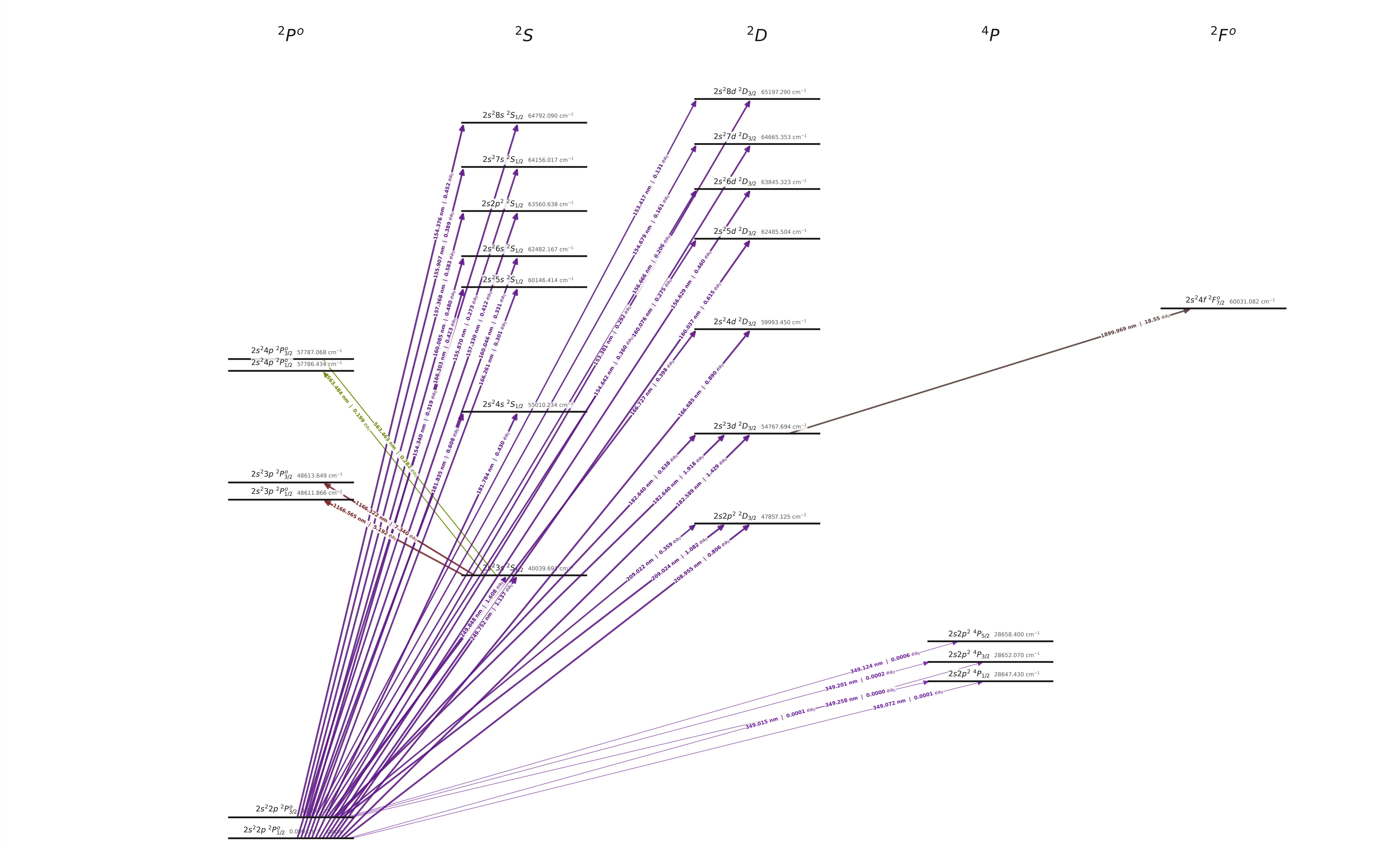


Boron

B (B I). Strongest classified transitions below 2 μm (NIST): 26 levels, 40 lines.

B⁺ ionisation limit 66928.04 cm⁻¹ = 8.29802 eV (149.414 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2s ² 2p 2P _{3/2} ^o - 2s ² 8s 2S _{1/2}	154.3763	-	1941.9597	0.452	5.620e+07	-	-	
2s ² 2p 2P _{3/2} ^o - 2s ² 7s 2S _{1/2}	155.9072	-	1922.8907	0.389	4.040e+07	-	-	
2s ² 2p 2P _{1/2} ^o - 2s2p ² 2S _{1/2}	157.3301	-	1905.5000	0.412	4.420e+07	-	-	
2s ² 2p 2P _{3/2} ^o - 2s2p ² 2S _{1/2}	157.3679	-	1905.0417	0.583	8.840e+07	-	-	
2s ² 2p 2P _{1/2} ^o - 2s ² 5d 2D _{3/2}	160.0371	-	1873.2683	0.615	4.670e+07	-	-	
2s ² 2p 2P _{3/2} ^o - 2s ² 6s 2S _{1/2}	160.0848	-	1872.7099	0.480	5.700e+07	-	-	
2s ² 2p 2P _{3/2} ^o - 2s ² 5s 2S _{1/2}	166.3032	-	1802.6858	0.423	3.940e+07	-	-	
2s ² 2p 2P _{1/2} ^o - 2s ² 4d 2D _{3/2}	166.6849	-	1798.5584	0.890	8.660e+07	-	-	
2s ² 2p 2P _{3/2} ^o - 2s ² 4s 2S _{1/2}	181.8349	-	1648.7070	0.608	6.220e+07	-	-	
2s ² 2p 2P _{1/2} ^o - 2s ² 3d 2D _{3/2}	182.5894	-	1641.8942	1.429	1.700e+08	-	-	
2s ² 2p 2P _{3/2} ^o - 2s ² 3d 2D _{5/2}	182.6399	-	1641.4402	1.918	2.040e+08	-	-	
2s ² 2p 2P _{3/2} ^o - 2s ² 3d 2D _{3/2}	182.6404	-	1641.4359	0.638	3.390e+07	-	-	
2s ² 2p 2P _{1/2} ^o - 2s2p ² 2D _{3/2}	208.8889	208.8889	1434.7205	0.806	3.610e+07	-	-	
2s ² 2p 2P _{3/2} ^o - 2s2p ² 2D _{5/2}	209.0235	208.9570	1434.2528	1.082	4.330e+07	-	-	
2s ² 2p 2P _{1/2} ^o - 2s ² 3s 2S _{1/2}	249.7522	249.6769	1200.3597	1.137	8.400e+07	-	-	
2s ² 2p 2P _{3/2} ^o - 2s ² 3s 2S _{1/2}	249.8476	249.7723	1199.9014	1.608	1.680e+08	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).